

Why Does Time Seem to Flow in Only One Direction?

A Resolution-Aware Program for Attributing the Thermodynamic Arrow

Anonymous Research Proposal

Abstract—The apparent direction of time connects several distinct facts: microscopic laws are often reversible, nonequilibrium processes produce entropy, macroscopic observers retain records of an apparently lower-entropy past, and cosmology confronts a low-gravitational-entropy initial-condition problem. Treating these as a single claim that entropy “makes time flow” conceals testable assumptions. This proposal develops the Resolution-Aware Arrow-of-Time Attribution Benchmark (RAAT), a computational program that compares forward–reverse path asymmetry, entropy-production accounting, fluctuation-relation consistency, record persistence, and resolution sensitivity across equilibrium, driven, hidden-state, and open-quantum model families. The central hypothesis is that an attributed dissipative arrow is coherent under valid resolution changes, protocol reversal, and nuisance controls, whereas preparation, measurement, or model-completeness artifacts fail at least one diagnostic. Cosmological content is kept as a boundary-condition ledger and toy-model comparator, not represented as a laboratory measurement. The work is strictly a research design: it specifies no executed simulations, no acquired trajectories, and no observed physical result.

Index Terms—arrow of time, entropy production, stochastic thermodynamics, fluctuation theorem, coarse graining, decoherence, gravitational entropy, model comparison

I. INTRODUCTION

Why does time seem to have a direction? Ordinary experience is asymmetric: glasses shatter but do not spontaneously reassemble, hot bodies cool in ordinary surroundings, and records are available of earlier rather than later events. Thermodynamics relates this macroscopic asymmetry to entropy increase, yet the statement needs careful scope. The underlying equations used in many microscopic models admit a time reversal operation, while the Second Law is a statistical and boundary-condition-sensitive statement about macroscopic descriptions. A rigorous answer must distinguish the arrow in a chosen system from a literal motion of time itself.

The distinction becomes sharper in cosmology. The early universe was hot and near equilibrium in its matter-radiation degrees of freedom, but its smoothness made it unusually low entropy with respect to gravitational clumping. Later structure formation and black-hole formation increase the available gravitational entropy. A claim that the Big Bang simply began in a high-entropy state is therefore incomplete. Equally incomplete is the claim that an eventual approach to equilibrium removes all microscopic temporal structure. A near-equilibrium system can still fluctuate, retain correlations for finite times, and be described by time-oriented boundary conditions; what disappears is a robust macroscopic thermo-

dynamic gradient, not the parameter used in a dynamical equation.

This proposal asks a tractable, evidence-bounded question: can finite trajectory data support an attribution of an observed temporal asymmetry to physical nonequilibrium dissipation rather than to an observation map, a preparation condition, a hidden variable, or a misspecified reversal operation? The proposed Resolution-Aware Arrow-of-Time Attribution Benchmark (RAAT) is a computational design with explicit negative controls. It does not promise an experimental solution of the cosmological initial-condition problem. Instead, it creates a common inference contract that reveals which part of a time-arrow claim is experimentally testable, which is conditional on coarse graining, and which remains a boundary-condition question.

II. THEORY SPINE AND EVIDENCE SURVEY

A. From reversible dynamics to an operational arrow

For a specified stochastic process, the direction of a sampled path may be quantified by comparing the probability of the path $x_{0:T}$ with the probability of its suitably reversed counterpart $\tilde{x}_{0:T}$. A natural path-asymmetry statistic is

$$\mathcal{A}[x_{0:T}] = \log \frac{p_\theta(x_{0:T} \mid \mathcal{O}, \Pi)}{p_\theta(\tilde{x}_{0:T} \mid \mathcal{O}, \Pi^R)}. \quad (1)$$

Here θ denotes the generator, $\mathcal{O}$ the observation map, Π the forward protocol, and Π^R a declared reverse protocol. Equation (1) is not intrinsically an entropy measurement. Its physical interpretation depends on local detailed balance, correct treatment of odd variables, a stated bath, and a reversal operation matched to the driving protocol.

Stochastic thermodynamics supplies this interpretation for a broad but delimited class of systems. It identifies work, heat, and entropy production at the trajectory level and derives fluctuation relations under declared assumptions [1]. Thermodynamic inference then confronts the practical difficulty that observers usually access only a subset of mesostates [2]. Landi and Paternostro show how entropy-production concepts extend across classical and quantum settings [3]; Ciliberto reviews representative experimental tests in mesoscopic systems [4]. Together, these sources support a direct claim: entropy production is quantifiable and falsifiable for specified models. They do not support the stronger claim that any temporal pattern, under any observation, is a measurement of physical dissipation.

For a finite-state continuous-time Markov model with stationary probabilities p_i and transition rates k_{ij} , the planned reference ledger is

$$\dot{S}_{\text{tot}} = \sum_{i,j} p_i k_{ij} \log \frac{p_i k_{ij}}{p_j k_{ji}} \geq 0. \quad (2)$$

Equation (2) defines the total entropy-production rate only after the state space and rates are declared. A partial observation can provide a useful lower-resolution asymmetry measure, but it does not automatically recover the complete physical ledger.

B. Coarse graining, records, and decoherence

Every laboratory description compresses degrees of freedom. Gell-Mann and Hartle describe quasiclassical coarse grainings based on approximately conserved quantities and connect information loss, decoherence, predictability, and thermodynamic entropy [5]. The important methodological consequence is not that entropy becomes arbitrary. Rather, a proposed entropy or arrow inference must declare the variables retained, the correlations discarded, and the temporal resolution used.

Records add another layer. Zurek analyzes how environmental coupling selects persistent pointer states and supports stable, approximately classical records [6]. Record persistence can thus help explain why an observer has reliable correlations with what is called the past. It must nevertheless be tracked separately from heat flow or total entropy production: a record statistic describes accessible correlations under a system-environment split, while Eq. (2) describes a thermodynamic ledger under a further set of assumptions. Strasberg and Winter provide a microscopic thermodynamic framework for open quantum systems and establish an appropriate fluctuation-theorem setting [9].

C. Cosmological arrow: boundary condition, not a shortcut

The cosmological case magnifies the issue of boundary conditions. A smooth early geometry contains less gravitational clumping than a universe filled with collapsed structures, motivating a low-gravitational-entropy past condition. Clifton, Ellis, and Tavakol propose a gravitational-entropy measure that increases with structure formation in selected cosmological geometries [7]. Their proposal is valuable precisely because it makes assumptions visible; it is not a universal, experimentally settled definition of gravitational entropy. Aguirre, Carroll, and Johnson analyze entropy-decreasing fluctuations in cosmological settings and show why reversals are conditional statistical histories rather than contradictions of all temporal ordering [8].

The report therefore reserves the phrase *cosmological boundary-condition ledger* for the low-entropy assumption required to orient a toy history. No simulation in RAAT measures the early universe, proves the past hypothesis, or predicts the far future. Rovelli’s perspectival suggestion is retained as a competing account of the relevance of coarse graining [11]; it is evaluated as a description-level alternative, not dismissed by definition.

III. SURVEY GAPS AND DIRECTION SELECTION

The Survey Agent identified six connected gaps. First, a forward–reverse score is frequently interpreted as uniquely physical even when model completeness is not documented. Second, partial observation and coarse graining are not compared by a standard resolution ladder across classical and quantum examples. Third, record formation, decoherence, and entropy production are often collapsed into a single narrative. Fourth, cosmological explanations may obscure their reliance on low gravitational entropy. Fifth, artifact controls for preparation and reversal conventions are not carried alongside an arrow claim. Sixth, the field benefits from public, auditable benchmark artifacts that retain these decisions.

The Idea Agent compared a universal entropy dashboard, a record-persistence program, a gravitational-entropy toy study, and an integrated attribution benchmark. It selected RAAT because it tests a sharp conditional hypothesis: a genuine, resolved nonequilibrium arrow should remain coherent under valid resolution changes, agree with its independent entropy ledger, survive symmetric protocol reversal, and outperform artifact models on held-out trajectories. A conclusion fails if the competing models are non-identifiable, if a valid coarsening reverses the interpretation, if nuisance injection mimics the signal, or if the reversal operation cannot be justified.

IV. RESEARCH DESIGN: RESOLUTION-AWARE ARROW ATTRIBUTION

A. Design objective and status

The ExperimentDesign Agent converts the selected idea into a computational design, not an executed study. Its unit of analysis is a finite trajectory ensemble paired with a fully declared model card. A model card records the generator family, state variables, system boundary, bath or environment assumptions, initial distribution, driving protocol, observation map, sampling resolution, reversal convention, permitted entropy ledger, and status of every trajectory. The only allowed trajectory status in this proposal is `SYNTHETIC_PLANNED`; no generated sample, fitted parameter, or score is reported as an observed result.

The design has two complementary objectives. The first is methodological: determine when a time-asymmetry statistic is interpretable as evidence of nonequilibrium dissipation. The second is explanatory: state exactly what remains outside that interpretation, including record formation, descriptive coarse graining, and the cosmological boundary condition. The design is ambitious because it connects classical, quantum, and cosmological language, but it is disciplined because it tests only the interfaces that have explicit generators and decision rules.

B. Generator families and negative controls

Table II defines five planned generator families. E0 is a stationary detailed-balance process. It is the critical negative control: under a symmetric observation map and a correctly matched reversal, it should not support persistent physical dissipation. E1 is a driven ring network or chemostatted

TABLE I
FOUR NONINTERCHANGEABLE COMPONENTS OF A TIME-ARROW CLAIM.

Component	Question it answers	Permitted evidence	Prohibited shortcut
Nonequilibrium dissipation	Does a declared system produce entropy under its stated dynamics?	Rates, bath model, driving protocol, and path statistics.	Treating every asymmetric series as dissipation.
Coarse graining	Which variables and time scales are retained by the description?	Observation map, aggregation, and sampling ledger.	Calling description dependence a proof that no physical arrow exists.
Records and decoherence	Why are some correlations stable and predictively useful?	System-environment split, record persistence, and decoherence model.	Equating record formation with total entropy production.
Cosmological condition	What boundary condition or gravitational-entropy proxy orients a toy cosmic history?	Explicit initial-condition and gravity-model assumptions.	Claiming that a finite benchmark measures the early universe.

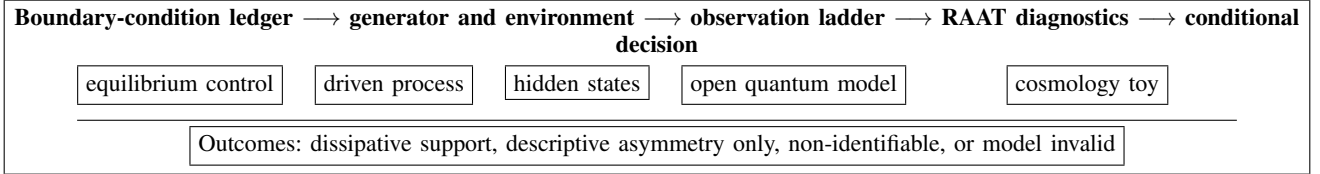


Fig. 1. RAAT attribution logic. The accompanying editable Draw.io source, `figures/multiscale_arrow_attribution.drawio`, retains the same separation between cosmological boundary conditions, physical generators, observation maps, diagnostics, and failure controls.

chemical-reaction network with a known affinity. Reaction-network thermodynamics provides a useful benchmark because its steady-state and transient contributions can be stated separately [12]. E2 hides or aggregates selected E1 states, exposing the difference between complete and partial observation. E3 uses a declared open-quantum dephasing or amplitude-damping model to test the system-environment and measurement-basis dependence of quantum arrow metrics. E4 is not a cosmological simulation campaign; it is a low-dimensional toy ledger whose sole purpose is to prevent a gravitational-entropy boundary parameter from being silently reclassified as an observed trajectory property.

The design does not assume that every family produces the same kind of entropy. Instead it asks whether the operational conclusion remains stable after the correct domain-specific conditions are supplied. For E0–E2, the rate ledger in Eq. (2) is available when transition rates and a suitable stochastic thermodynamic description are declared. For E3, an open-system entropy balance must be reviewed under the chosen channel and measurement assumptions [3], [9]. For E4, the only valid conclusion concerns the logical role of a boundary parameter in a toy model.

C. Resolution ladder and estimands

Let a latent trajectory be $z_{0:T}$, and let the reported trajectory at resolution r be

$$x_{0:T}^{(r)} = \mathcal{O}_r(z_{0:T}) + \varepsilon_r, \quad (3)$$

where $\mathcal{O}_r$ encodes state aggregation and sampling, and ε_r encodes a declared observation-noise model. The ladder changes one documented aspect of description at a time: sampling interval, state aggregation, accessible variable set, or measurement precision. It may reduce the recoverable entropy-production information. It may not be used to tune the answer after inspecting an outcome.

For N trajectories at resolution r , the planned mean directional score is

$$\bar{\mathcal{A}}_r = \frac{1}{N} \sum_{n=1}^N \log \frac{p_\theta(x_{0:T}^{(n,r)} | \Pi, \mathcal{O}_r)}{p_\theta(\tilde{x}_{0:T}^{(n,r)} | \Pi^R, \mathcal{O}_r)}. \quad (4)$$

The comparison of Eq. (4) with Eq. (2) is a consistency test, not an identity imposed by software. The design records an attribution discrepancy

$$D_r = \left| \bar{\mathcal{A}}_r - \hat{\Sigma}_{r,\text{ledger}} \right|, \quad (5)$$

where $\hat{\Sigma}_{r,\text{ledger}}$ is defined only for a permitted ledger at the stated resolution. In a hidden-state condition, the complete-model ledger and the partial-observation estimator are stored separately. This prevents an attractive but false conclusion that a lower-resolution value has directly measured every unobserved contribution.

Where an applicable fluctuation relation is available, the proposed residual is

$$R_r(s) = \log \frac{P_r(s)}{P_r(-s)} - s, \quad (6)$$

with s the predeclared entropy-like trajectory statistic and P_r its resolution-specific distribution. Equation (6) is diagnostic only under its theorem’s conditions. It is not applied to a cosmological toy proxy or to an arbitrary series merely because the series appears irreversible. Carberry et al. motivate the value of transient fluctuation-theorem tests in a controlled colloidal setting [10]; RAAT transfers the logic of matched controls, not an unearned claim of experimental replication.

D. Inference protocol and decision contract

Each planned condition has a training and held-out trajectory partition, a fixed candidate-model set, and a nuisance

TABLE II
PLANNED RAAT GENERATOR FAMILIES AND FALSIFICATION CONTROLS. NO FAMILY IS EXECUTED BY THIS PROPOSAL.

ID	Role	Declared latent truth	Required falsification control
E0	Detailed-balance negative control	Stationary zero steady-state entropy production.	Symmetric sampling and reversal; an apparent arrow indicates estimation or protocol failure.
E1	Driven stochastic positive control	Nonzero affinity and an independently computed entropy ledger.	Reverse the protocol and perturb rates, temperature, and initial distribution.
E2	Partial-observation stress test	Same underlying driven generator as E1, with hidden or merged states.	Compare each observation map with the complete-state reference; never equate them by fiat.
E3	Open-quantum bridge	Declared channel, basis, bath, and system-environment split.	Vary the measurement basis and environment model; reject an interpretation that is partition dependent without disclosure.
E4	Cosmological toy comparator	Explicit low-gravitational-entropy proxy and boundary ledger only.	Disallow claims about measured early-universe entropy or a unique gravity entropy.

suite. The proposed likelihood is

$$\mathcal{L}(\theta, \eta) = \prod_{n=1}^N p_{\theta, \eta} \left(x_{0:T}^{(n)} \mid \Pi, \mathcal{O}_r \right) \prod_k \pi_k(\eta_k), \quad (7)$$

where η contains nuisance parameters such as observation noise, hidden-state fraction, protocol-time offset, or temperature mismatch, and π_k are their declared constraints. Candidate explanations are compared by held-out predictive support, the stability of $\bar{\mathcal{A}}_r$, the discrepancy in Eq. (5), the residual in Eq. (6), and whether a nuisance configuration reproduces the apparent direction.

The key output is a conditional classification rather than a single dramatic number. A condition receives `DISSIPATIVE_SUPPORT` only if all four requirements are met: (i) the sign and size of the relevant score are stable across a prespecified valid segment of the resolution ladder; (ii) the score is compatible with the independently declared entropy ledger; (iii) a protocol-symmetric reversal does not remove the conclusion; and (iv) the nearest artifact model loses predictive support on held-out trajectories. It receives `DESCRIPTIVE_ASYMMETRY_ONLY` when a record or coarse variable is directionally predictive but complete physical attribution cannot be established. It receives `NON_IDENTIFIABLE` when competing mechanisms are observationally equivalent. It receives `MODEL_INVALID` when the system boundary, reversal map, or relevant theorem conditions cannot be justified.

V. PLANNED EVALUATION AND EXPECTED OUTCOME BRANCHES

A. Predeclared analyses

For every family, the planned analysis uses paired bootstrap intervals over trajectories, with generator-level and observation-level perturbations retained as distinct uncertainty sources. The design reports the entire resolution profile rather than a selected best resolution. A calibration phase uses E0 and E1 to check that the implementation of time reversal, protocol reversal, and entropy accounting has the expected logical behavior. It does not use calibration to claim a physical discovery. The E2 family then asks whether an apparent arrow

TABLE III
CONDITIONAL CONCLUSION BRANCHES SPECIFIED BEFORE EXECUTION.

Planned pattern	Permitted conclusion
Stable score, ledger agreement, reversal control, and held-out discrimination	Support for dissipative attribution in the declared model and resolution range.
Stable directional records but incomplete ledger or unresolved hidden states	Descriptive asymmetry is supported; physical dissipation is not fully attributed.
Similar predictive support for driven and artifact models	Mechanism is non-identifiable from the available variables.
Failure of symmetry, local detailed balance, or reversal assumptions	Model invalid for the intended thermodynamic conclusion.
Toy boundary parameter required to orient a cosmological history	Boundary-condition dependence is exposed; no empirical measurement of the early universe follows.

is reduced, distorted, or non-identifiable after hidden variables are introduced. E3 tests whether a conclusion is robust to an explicitly changed measurement basis or environment partition. E4 exposes where a cosmological analogy requires an initial-condition input instead of a measured dissipative trajectory.

The primary comparison is not “forward score greater than zero.” It is a four-way contrast between a detailed-balance explanation, a driven-dissipative explanation, a partial-observation explanation, and a protocol or measurement artifact explanation. The analysis logs which assumptions distinguish these candidates. This is important because time-arrow discourse often grants an interpretation to a statistic before the statistic’s alternative generative stories have been tested.

B. What a successful result would and would not establish

If the planned benchmark finds an admissible regime of `DISSIPATIVE_SUPPORT`, the correct conclusion is narrow but valuable: in that model family, at that observation res-

olution, and under that reversal convention, the data pattern is better explained by the declared nonequilibrium entropy-production mechanism than by the tested alternatives. This would support a scientific link between macroscopic irreversibility and trajectory-level asymmetry. It would not show that time itself is generated by entropy, establish a universal entropy function for gravity, or prove why the universe began in a special state.

Conversely, a `DESCRIPTIVE_ASYMMETRY_ONLY` outcome is not a failure. It would quantify how record-bearing and coarse descriptions can carry a useful temporal orientation while not licensing complete thermodynamic attribution. A `NON_IDENTIFIABLE` outcome is also informative: it identifies what new variables, observation precision, or theoretical structure would be needed. These branches avoid the common practice of interpreting a null or ambiguous result as evidence for a preferred philosophical narrative.

C. Relation to cosmological claims

The benchmark’s cosmology module is deliberately asymmetrical in ambition. It can test bookkeeping questions: whether a toy history has a stated low-entropy boundary, whether a proposed coarse-grained entropy proxy rises under selected dynamics, and whether a conclusion depends on the proxy. It cannot decide the origin of the boundary condition. Gravitational entropy remains an active theoretical topic, and an operational measure suitable for a black hole or a perturbed cosmological geometry does not automatically yield a universal answer [7]. This is not a defect of the design; it is the point of the boundary ledger. It tells the reader where a thermodynamic explanation ends and where a cosmological assumption begins.

VI. THEORY-TO-OBSERVATION BRIDGE

A. A hierarchy of claims

The phrase “the arrow of time” can name claims with very different evidentiary status. At the strongest empirical level, a controlled finite system can display a time-asymmetric trajectory distribution that is consistent with a specified entropy-production ledger. At an intermediate descriptive level, an observer can have stable records and predictive variables even though the complete thermodynamic ledger is inaccessible. At a cosmological level, a model can make its low-entropy boundary condition explicit and investigate the consequences of that assumption. These levels should communicate, but none should be promoted to another level merely because they share the word entropy.

RAAT uses this hierarchy to prevent a common inferential error. Suppose that a process produces a large positive value of Eq. (4). That value is direct evidence only for the asymmetry of the modeled forward and reverse path distributions. To call it physical dissipation, the state variables, bath, work protocol, and reversal operation must be justified. To connect it to a record, a persistence criterion and a system-environment split must be provided. To connect it to cosmology, an additional initial-condition and gravity model must be supplied. The

proposed workflow records every transition in this chain as an assumption-bearing interface.

This does not make the explanation weak. It makes it discriminating. A thermodynamic account that survives the interfaces identifies the circumstances in which macroscopic irreversibility is more than a visual asymmetry. A record account that survives its interface identifies why a finite observer has useful temporal correlations. A cosmological account that states its boundary condition identifies the exact explanatory burden that remains. Separating these achievements makes it possible for future evidence to strengthen one link without claiming to have settled all links.

B. Protocol symmetry and causal alternatives

Time reversal in a physical model is not equivalent to reading a stored sequence backwards. Velocities, magnetic fields, externally driven controls, and measurement operations can have different parity or protocol transformation rules. A benchmark that reverses an array while leaving a time-dependent drive unchanged can create an artificial directional contrast. RAAT therefore makes protocol symmetry a first-class object. For every planned forward condition, the design specifies how Π^R changes the drive, which state variables are odd under reversal, and which data transformations are merely display operations.

The same distinction matters for causality. A record correlated with an earlier state does not by itself prove that the earlier state caused the record under every intervention. In the proposed benchmark, causal language is reserved for generator-level interventions: changing an affinity, changing a bath parameter, hiding a state, or introducing a measurement asymmetry. Each intervention has a predeclared alternative explanation and a response pattern. The analysis then asks whether the observed temporal score changes in the direction predicted by the model card and whether an artifact injection can reproduce that change. This is stricter than narrating a direction from an unmanipulated series.

For a candidate model M , the proposed cross-resolution stability functional is

$$\mathcal{S}(M) = \sum_{r \in \mathcal{R}_{\text{valid}}} w_r \mathbf{1} \left[\text{sign}(\overline{\mathcal{A}}_r) = \text{sign}(\widehat{\Sigma}_{r, \text{ledger}}) \right] - \lambda \overline{D}_r, \quad (8)$$

where $\mathcal{R}_{\text{valid}}$ is the prespecified valid portion of the observation ladder, w_r are declared resolution weights, $\overline{D}_r$ is the planned aggregate discrepancy from Eq. (5), and λ is fixed before analysis. Equation (8) is not a universal entropy measure. It is a transparent scoring aid for deciding whether a particular attribution is stable enough to merit the stronger conclusion branch.

C. Records, memory, and the direction of inference

The familiar subjective sense that the past is fixed while the future is open also has an information-theoretic component. Records, traces, and memories are present variables correlated with selected earlier variables; they are physical systems and therefore carry costs, noise, and a finite lifetime. Decoherence

can make such records robust by dispersing phase information into an environment [6]. Coarse graining can make the resulting description predictable by suppressing inaccessible detail [5]. Neither observation establishes that an observer’s access to records is identical to the total entropy increase of a closed universe.

RAAT operationalizes this distinction with a planned record-persistence statistic. A record variable is eligible only if its correlation with a declared target remains above a prespecified threshold across a fixed forecast horizon and if the noise model is documented. The statistic is reported next to, not inside, the entropy ledger. This architecture allows several informative patterns. A driven system may exhibit both record persistence and dissipative support. A partially observed system may retain directional records while receiving only a descriptive-asymmetry classification. An equilibrium negative control may produce transient correlations without supporting a persistent net-dissipation attribution. These distinctions offer a more precise answer to the popular question than an all-purpose slogan about disorder.

D. Reproducibility as an explanatory control

The benchmark’s final bridge is reproducibility. In foundational work, an undocumented reversal convention or an unrecorded aggregation choice can change the apparent philosophical conclusion. RAAT therefore treats artifacts as part of the evidence: generator equations, parameter cards, initial-state distributions, observation maps, reversal functions, seeds, inference versions, and outcome-class decisions must be retained together. The same principle constrains use of cosmological analogies. A toy low-gravitational-entropy proxy must list its gravity assumptions and may not be rendered in a plot with the visual authority of an observed cosmological quantity.

This documentation policy is also a route to productive disagreement. Different researchers may favor different coarse grainings, quantum partitions, or gravitational-entropy proposals. Rather than forcing agreement through a single score, the benchmark exposes how the conclusion changes under each declared choice. A result that is stable across legitimate alternatives earns broader support. A result that is not stable identifies the missing physics or measurement needed for resolution. In both cases, the research program turns a diffuse debate about time into a set of transparent, falsifiable interfaces.

VII. RISKS, LIMITATIONS, AND HUMAN REVIEW

The largest technical risk is overinterpreting a coarse-grained trajectory statistic. Hidden variables can reduce or distort apparent entropy production, and nonstationary preparation can create directional signatures even when the intended stationary analysis is inapplicable. The design mitigates this by storing observation maps, using E0 and E2 controls, and allowing `MODEL_INVALID` rather than forcing a scalar result into a physical interpretation. A second risk is treating a theorem as portable across all settings. The fluctuation residual

in Eq. (6) is evaluated only when its assumptions have been independently checked.

The quantum bridge introduces an additional partition risk. The definition of heat, work, and entropy can depend on the chosen system-environment split, basis, and measurement protocol. The design therefore requires an open-quantum-systems review before any E3 outcome is interpreted. The cosmological comparison carries a stronger limitation: it cannot reproduce the early universe or settle a unique gravitational entropy. A cosmologist must review the E4 proxy, its relationship to the Weyl-curvature or other gravitational-entropy ideas, and every sentence that connects a toy result to cosmology.

Finally, reproducibility is a scientific requirement, not merely a software preference. Every planned run must retain generator source, parameter card, seed policy, observation map, protocol, inference code version, and decision thresholds. Synthetic records must remain labeled as synthetic. This design does not authorize automatic experimental control, does not operate a physical device, and does not assert a result before those artifacts exist and are reviewed.

APPENDIX A

EVIDENCE COVERAGE AND DIRECTION SELECTION AUDIT

The Survey Agent constrained the proposal to twelve registered sources. Cross-validated entries establish the main formal and review background: stochastic thermodynamics and fluctuation theorems [1], thermodynamic inference under partial observation [2], quantum-to-classical entropy-production perspectives [3], and experimental context [4]. Gell-Mann and Hartle support the treatment of coarse graining and quasiclassical descriptions [5]; Zurek supports the separate accounting of records and decoherence [6]. Clifton, Ellis, and Tavakol anchor the gravitational-entropy discussion [7].

Three registry entries were discovered through OpenAlex but await publisher-page verification because the user-requested in-app browser could not start in the Windows ACL sandbox: the fluctuation analysis of cosmological lower-entropy histories [8], the open-quantum first- and second-law framework [9], and the colloidal fluctuation-theorem demonstration [10]. Rovelli’s coarse-graining perspective [11] is likewise retained as a registered alternative rather than used as an unsupported assertion. These verification statuses are recorded in the Survey manifest; they do not turn metadata discovery into source-page validation.

The simulated Idea Agent initially considered a monolithic entropy dashboard. It rejected that route because it would conflate G1 operational attribution, G2 partial observation, G3 records, and G4 cosmological boundary conditions. It selected RAAT after requiring a common model card, a resolution ladder, competing generative models, and explicit falsifiers. The ExperimentDesign Agent then enforced a computational `DESIGN_ONLY` policy, an empty observed-result set, and human review for stochastic, quantum, and cosmological interpretations.

TABLE IV
SYMBOLS USED IN THE PROPOSED INFERENCE PROTOCOL.

Symbol	Definition
$x_{0:T}$ and $\tilde{x}_{0:T}$	Observed trajectory and declared time-reversed trajectory.
$z_{0:T}$	Latent trajectory before observation or aggregation.
θ and η	Generator parameters and nuisance parameters.
Π and Π^R	Forward and explicitly matched reverse protocols.
$\mathcal{O}_r$	Observation map at resolution level r .
$\mathcal{A}$	Forward–reverse path likelihood asymmetry.
$\dot{S}_{\text{tot}}$	Model-specific total entropy-production rate.
$R_r(s)$	Applicable fluctuation-relation residual.

APPENDIX B SYMBOLS AND OPERATIONAL DEFINITIONS

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